

## Yet Another Simple Pot-controlled 555 PWM generator

by [ZeroTruths](#) on July 10, 2010

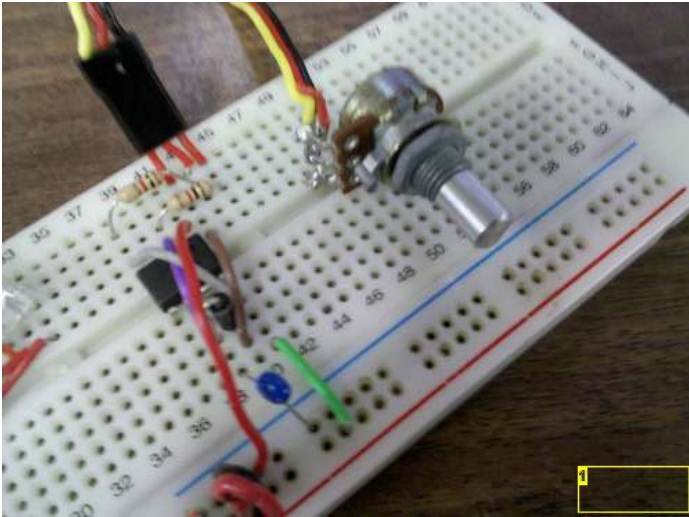
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## Intro: Yet Another Simple Pot-controlled 555 PWM generator

Something that a project that I'm working on has me doing is using a serial to parallel IC (think 74HC595) to control leds. However, rather than drive the leds directly from the pins, I opted for the use of transistors. After testing this out, it became apparent to me that perhaps the leds might be too bright, so I went in search of a simple PWM generator.

Of course, there are a couple of instructables that already feature such a circuit, but I was unable to get them to work correctly for whatever reason. This being the case, I will now present the circuit that I came across and works very well.



### Image Notes

1. What you'll be building!

## Step 1: Yet another overview

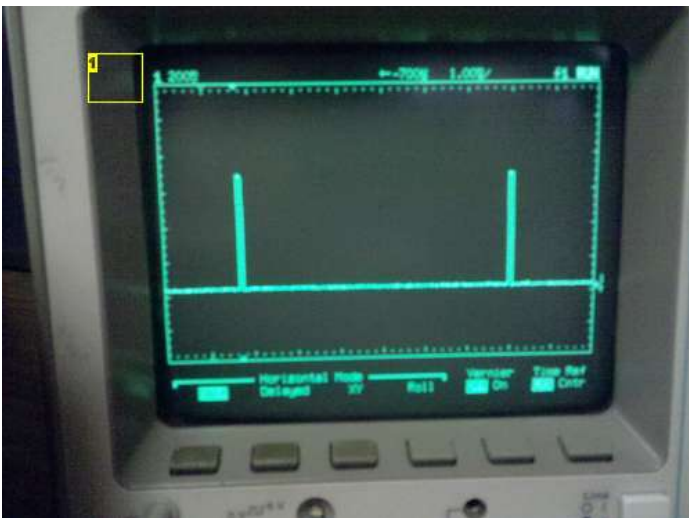
**PWM** stands for **P**ulse **W**idth **M**odulation, which is a simple way of efficiently supplying varying amounts of power.

For example. Say that you wanted to control the brightness of an led (note: there are many ways to do this, but for the sakes of an example, I'll only note two). The first way would be to put a variable resistor in series with the led. This would alter the amount of current that went through the led, while holding the voltage constant. If you put the variable resistor at 40%, the led would be 40% as bright as it could be.

The second way would be to connect a led in series with a resistor and a power supply that could be turned on and off *really* quickly. Let's say that you were able to turn on and off the power supply quick enough to the point where 40% of the time it was on, and 60% of the time it was off. This would be reflected by the led being on full brightness, but only for 40% of the time, giving the illusion of being 40% as bright as it could be.

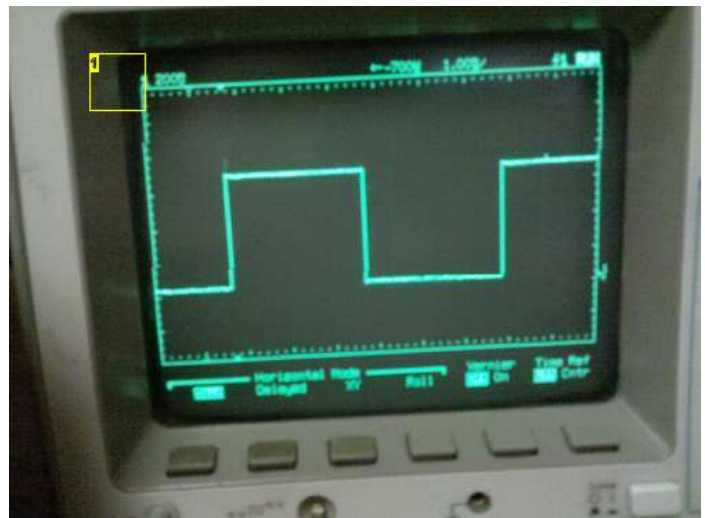
Two different methods, for the same result. What's the difference? About 60% of the energy gets burned off as heat in the resistor in the first circuit, while in the second circuit, almost all of the energy supplied is used.

This is why PWM is useful. It allows a signal to range from completely off to completely on. If a signal is turned on and off quick enough, given a certain ratio, a signal can appear to be that ratio, without suffering from much power loss at all.



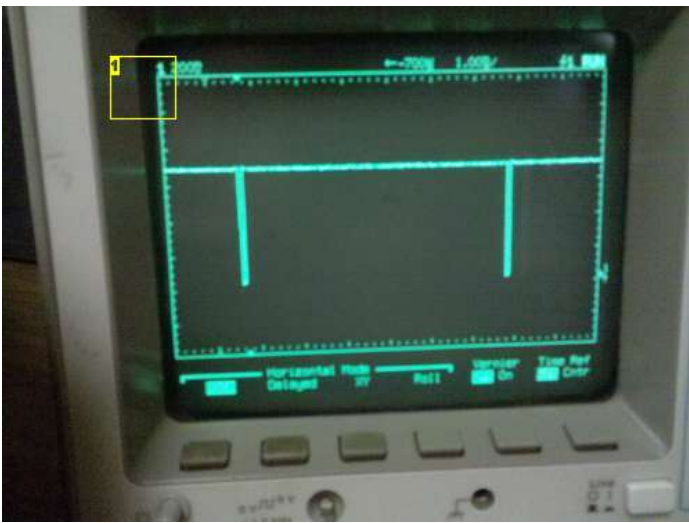
### Image Notes

1. PWM @ 0%



### Image Notes

1. PWM @ 50%



#### Image Notes

1. PWM @ 100%

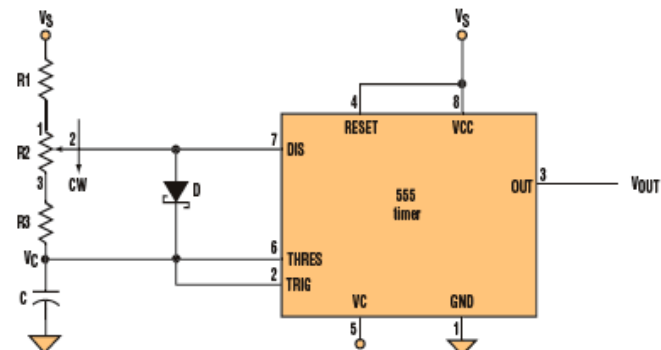
### Step 2: What will be needed

The schematic that this circuit is based on is so general, that instead of giving hard values for each component, the author gave relations that would allow for any combination of parts.

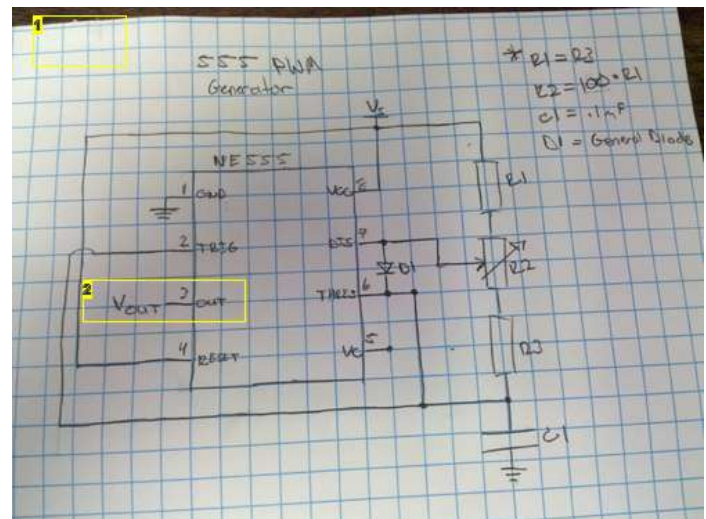
However, for the sake of getting it right the first time, I'll list the values that did work.

#### Components:

- \* 1 - Generic 555 IC (NE555 was used)
- \* 2 - 1K Resistors (R1, R3)
- \* 1 - 100K Linear Potentiometer (  $R2 = 100 \cdot R1$  )
- \* 1 - 1n4004 Diode (Pretty much any diode will do)
- \* 1 - .1uF Ceramic Capacitor (Unsure about the relation of the value of this to the resistors)
- \* Breadboard



1. Adding a diode and potentiometer to this 555 timer circuit results in a wide-range PWM.



#### Image Notes

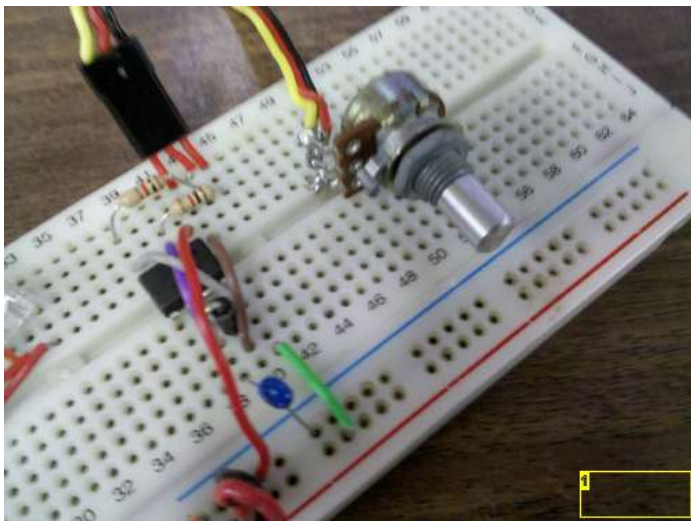
1. Hand drawn view
2. Output signal

### Step 3: Build it!

There's nothing much else to this circuit except for just going for it and building it.

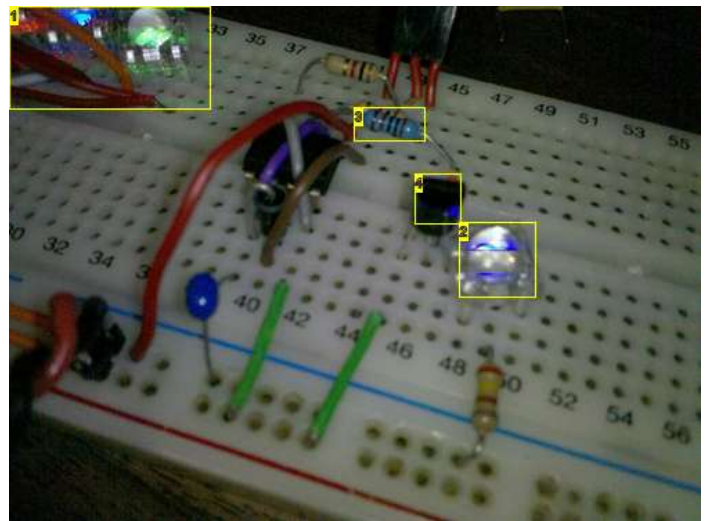
Know of the proverb "Measure twice, cut once"? The same thing applies. Build the circuit, check the circuit, *then* apply the power.

Using the circuit is simple! Your PWM signal will be coming from pin 3. From there, make the standard led circuit, except route the voltage to pin 3. Play with the pot and enjoy!



#### Image Notes

1. What you'll be building!



#### Image Notes

1. Another project
2. It's on, but turning on and off really quickly
3. Using the PWM as a signal, so lets limit the current for the transistor
4. Using an NPN transistor as a gate, controlled by the signal

## Step 4: Credits / Sources

Article discussing the schematic: <http://electronicdesign.com/article/analog-and-mixed-signal/wide-range-pulse-width-modulator-uses-555-timer110.aspx>

Schematic: [http://archive.electronicdesign.com/files/29/11028/figure\\_01.gif](http://archive.electronicdesign.com/files/29/11028/figure_01.gif)

## Related Instructables



**Simple controllers for DC motors (inc PWM) inertia and braking (sim)** by marshon



**Very simple PWM with 555...Modulate every thing** by shams



**learn about the 555** by Jimmy Proton



**Simple Servo Tester** by nmcciana



**Tv-B-Gone Hat** by m\_jake



**Simple (and dirty) Pulse Width Modulation (PWM) With 555 Timer** by Chein

## Comments

7 comments

[Add Comment](#)



**ledfreak14** says:

Did you know you can play tetris on that oscilloscope?

Feb 25, 2011. 6:09 AM [REPLY](#)



**rcisneros** says:

Thanks for the post.

I know it's my own ignorance, but there is not info here for me. The levels of the readers electronics knowledge varies greatly.

Like for me, I don't see how this even works since I can't find where to attach the power, where it comes out. I have to figure out ground, might be the (-). I know it's me, but help a guy out and label those things.

Keep in mind that the people that can read this cold, probably know how to do it already.

Thanks

Oct 15, 2010. 11:21 PM [REPLY](#)



**ZeroTruths** says:

Thanks for the reminder. It's true that I do often overlook this.

The 555 IC can take a voltage of between 5-15v, which means that you could power it with a simple 9v battery. In schematics, Vs stands for Voltage Source, which is where you would plug the positive (+) terminal of the voltage source, while the triangle (in the first picture in this step) is where you would connect the negative (-) terminal.

Schematics that have multiple ground symbols in actuality only have one. When creating the circuit, all of those ground symbols will connect together and meet at the battery's negative terminal.

The output voltage, where the signal is generated, comes from pin 3 on the IC.

Oct 19, 2010. 12:27 PM [REPLY](#)

Remember, all grounds meet together, unless otherwise specified.

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**rimar2000** says:

Jul 11, 2010. 7:46 AM **REPLY**

Good work! Now, a question: can I use this PWM to vary the velocity of a desk fan? And a drill? In other words: how many potency can this circuit manage?

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**ZeroTruths** says:

Jul 11, 2010. 11:18 AM **REPLY**

Directly? No. If you were to directly connect the output pin to the device, the most current you'd get would be something around 30mA, I believe. However, that's still not to say that it couldn't be done with this circuit. What you could instead do is use the output of this circuit and use it to control the gate pin on an NPN transistor, similar to what I did on the very last image. You'd have to make sure that the transistor could take the kind of currents that you need (drills and fans tend to use a lot of current, around the range of Amps)

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**rimar2000** says:

Jul 11, 2010. 2:34 PM **REPLY**

Thanks very much for your response. Electronic, as you can see, isn't one of my skills...

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**omnibot** says:

Jul 11, 2010. 1:13 AM **REPLY**

Nice one.

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